

APAC Technology

2026 OCP APAC Summit: AI Infrastructure a Full-Stack Systems Race

OCP APAC Summit 2026: Power, Packaging and Networking key the build-out

In conjunction with our Taiwan semi team (Sunny Lin), we attended the OCP APAC Summit in Taipei Aug 10-11, the industry forum led by the leading hyperscalers to design data center infrastructure around efficiency, scalability, and open standards. We attended the exhibition and presentations across the keynotes, power electronics, networking and semiconductors/advanced packaging. The report includes session takeaways from AMD, Applied Materials, Arm, ASE, Broadcom, Google, Microsoft, NVIDIA, TSMC, Delta, Hon Hai, Lite-On, Renesas, Vertiv, Astera Labs, and an update on ASpeed. The common message was that AI infrastructure is moving toward a full-system approach to advancing and optimizing across compute, memory, networking, power, cooling, security, packaging, storage, and rack/data-center architecture.

Power electronics: 400V/800VDC still locked in as the primary upgrade path

Power presentations across Microsoft, Google, NVIDIA, Delta, Hon Hai, Lite-On, Renesas, and Vertiv highlighted the migration of the power electronics into separate power side cars from 2027 to drive 400/800V into the IT rack. Our feedback points to ~10% penetration from Rubin and advancing further on Rubin Ultra even with the Kyber delay as compute wattage further increases but at a faster pace (30-40%) on upcoming 400V+/- ASIC systems from Google and Meta. The upgrade to HVDC enables benefits of increasing rack density for compute while reducing copper, cable bulk, conversion losses, heat, and rack space consumed by power infrastructure, enabling multi-hundred-kilowatt and eventually megawatt-class AI racks. Participants also are developing power cell and power block facility and row level rectifiers noted in our Computex [report](#) and solid state transistor starting now in China ([link](#)) and outside China from 2028-30 to meet safety certification standards.

Advanced packaging: scaling shifts from transistors to system integration

TSMC, ASE, Applied Materials, and Broadcom reinforced that AI performance increasingly depends on advanced packaging using the toolbox including CoWoS and migration toward CoPoS, SolC, fan-out/FoCoS, CPO, and system-technology co-optimization. The key bottlenecks are no longer just leading-edge logic nodes, but yield, warpage, substrates, memory availability, thermal design, power delivery, optical alignment, and ecosystem-level qualification. TSMC continues to ramp advanced packaging with excellent yields and promote its 3DFabric for advanced silicon integration while ASE discussed its VIPack covering fan-out, 2.5-3D, FoCoS, and CPO.

Networking and optics: data movement the next scaling bottleneck

CPO is emerging as a key solution to AI networking bottlenecks as copper reach, pluggable reliability, and chip-edge bandwidth constraints worsen at higher speeds. Moving optics closer to ASICs and accelerators could improve power efficiency, reliability, training uptime, and scale-up/scale-out flexibility across larger AI systems. The AI interconnect value chain may shift toward CPO, optical I/O, advanced packaging, and open standards that reduce power, cost, and vendor lock-in. ASE is positioning CPO as a natural extension of advanced packaging, moving optics, ASICs, switches, HBM, and power delivery into tighter system-level integration for AI servers.

Compute stack: agentic AI broadens the opportunity beyond GPUs.

AMD and Arm both emphasized that agentic AI increases demand for CPU orchestration of memory, storage, RAG/vector databases, tool execution, and system-management workloads alongside GPUs. The company's expressed see roles including ARM's new

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AGI processor and AMD seeing strong demand ramping up its new Venice CPU. Microsoft, Google, and NVIDIA framed AI leadership for agentic AI as vertical integration from silicon to racks and data-center operations, with custom silicon and heterogeneous compute used to optimize tokens per dollar, performance per watt, supply availability, and fleet utilization. At the hardware level, NVIDIA's MGX, Google's Deschutes/Brazos cooling, Hon Hai's liquid-native pod architecture, Microsoft's Mt. Diablo validation work, and Vertiv's sidecar/power-center roadmap all point to racks becoming modular infrastructure platforms. Compute, power, liquid cooling, telemetry, BMS/DCIM, workload scheduling, leak detection, safety, and serviceability need to be designed together to keep expensive accelerators highly utilized.

Stock implications: AI capex broadens into the full infrastructure value chain.

OCP showcased the industry push to drive more compute and build architectures to improve performance and efficiency across the scale-in, scale-out and scale-across domains, with the full stack approach driving a diversified infrastructure investment across compute, storage, networking and power. Our preferred semiconductor beneficiaries are TSMC, ASE, ASpeed, and MediaTek, driven by advanced packaging, CPO/optical packaging, BMC/security, custom silicon, and system-level integration. In hardware, key beneficiaries include server ODMs Hon Hai, Quanta, Wistron, and Wiyynn, while component and infrastructure names such as Delta, King Slide, and BizLink should benefit from higher rack density, 800VDC power, liquid cooling, cabling/interconnect, mechanical content, and AI rack buildouts.

PANEL & KEY THEME HIGHLIGHTS

We attended the main keynote track, power electronics track, and semiconductor/advanced packaging presentations at OCP APAC Summit 2026 in Taipei, including sessions from AMD, Applied Materials, Arm, ASE, Broadcom, Google, Microsoft, NVIDIA, TSMC, Delta, Hon Hai, Lite-On, Renesas, Vertiv, Astera Labs, and ASpeed. The common message was that AI infrastructure is moving beyond GPU procurement toward full-system optimization across compute, memory, networking, power, cooling, security, packaging, storage, and rack/data-center architecture.

About the Open Compute Project (OCP)

The OCP is an industry consortium (over 400 members as of 2025) launched by Facebook (now Meta) in 2011 to redesign hardware technology for data centers with a focus on efficiency, scalability, and openness. It promotes open-source hardware designs for servers, storage, networking, and data center infrastructure, enabling hyperscalers and enterprises to standardize, reduce costs, and accelerate innovation by sharing blueprints and collaborating across vendors. The standards have been a key enabler of Taiwan's 'whitebox' model supporting the server and networking ODMs along with more opportunity for silicon, packaging, power supply and cooling.

Keynote track (Randy Abrams)

The 2026 OCP APAC Summit in Taipei opened with keynotes from AMD, Applied Materials, Arm, ASE, Broadcom, Google, Microsoft, NVIDIA, and TSMC which framed AI infrastructure as a full-stack systems problem rather than a GPU-only procurement cycle. AMD and Arm emphasized that agentic AI expands the CPU, memory, orchestration, and rack-level opportunity; Microsoft, Google, and NVIDIA positioned AI infrastructure leadership around vertical integration from silicon to data-center operations; while Applied Materials, ASE, Broadcom, and TSMC focused on the enabling technologies—advanced packaging, 3D integration, optics, power, and system-level co-optimization—that make the next scaling wave possible.

The speakers highlighted that AI performance is shifting beyond raw accelerator performance toward data movement, power delivery, cooling, and reliability at scale. Open standards such as OCP, ESUN, OCI, MRC, UALink, CXL, Caliptra, and modular rack specifications are gaining importance as hyperscalers push for interoperable, multi-vendor infrastructure. At the same time, the industry is moving toward CPO and silicon photonics for lower-power networking, 400V/800VDC power architectures for higher rack density, liquid cooling for thermal headroom, and CoWoS/SolC/3.5D packaging to bring compute, HBM, cache, and I/O closer together.

The conference reinforced that AI investments are broadening beyond the accelerators into the infrastructure required to keep accelerators utilized. The supply chain opportunities will continue to advance across server CPUs, networking silicon, optics/CPO, advanced packaging, substrates, HBM integration, power electronics, 800VDC distribution, BBUs/BESS, cooling, telemetry, and rack-level integration.

AMD: Building Open Data Centers for Agentic AI: A Blueprint for AI at Scale

Speaker: Ravi Kuppuswami, SVP of Server Product and Engineering

Key message: AMD (covered by UBS analyst Tim Arcuri) views Agentic AI expanding the infrastructure opportunity beyond GPUs by driving more CPU compute, memory, networking, management, power, and cooling at the rack and data-center level. The shift toward more balanced CPU/GPU architectures and standardized OCP racks/fabrics could increase demand for server CPUs, networking, rack integration, and modular infrastructure. AI capex may broaden from accelerator scale-up into full data-center system build-outs enabled by open standards.

- **Agentic AI is a rack- and data-center-level problem.** AI infrastructure now needs to integrate workflows, silicon, firmware and trust, fabric and memory, rack architecture, power, and cooling. Memory is also critical—anyone not thinking deeply about memory is at a disadvantage today.

- **Agentic AI is bringing more CPU compute to the forefront.** The traditional data center was built around training and chatbot workloads, with x86 servers handling web, caching, applications, databases, and storage alongside eight-way GPU servers. Agentic infrastructure is now driving more x86 servers for web, applications, databases, and storage; new agentic CPU racks for the agent control plane and tool execution; and continued rack-scale GPU deployment. This does not reduce the need for rack-scale GPUs, but it adds significantly more CPU compute to the floor. A traditional CPU-to-GPU ratio for inference was 1:4, while the new infrastructure is shifting toward 1:1.
- **CPU racks are central to agentic AI workloads.** CPUs provide compute for the agent gateway, planning, orchestration and scheduling, RAG/vector databases for data retrieval, reasoning, KV cache, tool execution and verification, and response generation. CPUs are also becoming distributed across different systems, including the web gateway, agentic orchestration layer, SaaS/IaaS platforms, GPU/ASIC racks, distributed KV cache, and storage/knowledge services. OCP is driving standards across the data center (UALink, UEC, ESUN, USLoE), rack (ORv3, ORW), platform (DC-MHS, OAC), firmware/security (Redfish, Caliptra), and components (DC-SCM, OCP NIC, DAC card).
- **Infrastructure relies on manageability, fabric, and acceleration.** The core building blocks include the DC-SCM BMC module, OCP NIC for networking, and accelerator cards. For firmware, security, and manageability, the architecture uses a Redfish management interface, Caliptra root of trust, openSIL firmware to abstract silicon complexity, and OCP SAFE for security assurance. To support diverse configurations, it also needs a modular server architecture for standardized accelerator integration, including open accelerator cards (OAC) and DC-MHS compute and storage platforms.
- **Standard racks enable deployment of both CPUs and accelerators.** ORW extends ORv3 into a double-wide format for training and inference workflows in agentic data centers. ORv3 provides diverse rack profiles for different agentic use cases, while improving flexibility, and choice across rack deployments. Scale-up standards such as UALink, UALoE, and ESUN help a rack behave like a giant accelerator. Scale-out standards such as UEC deliver congestion control and low latency for AI cluster traffic between racks and pods.

Applied Materials: Accelerating energy-efficient computing through Systems Technology Co-Optimization (STCO)

Speaker: Subi Kengeri, corporate VP and GM, Systems and Materials

Key message: AI demand remains a strong secular driver, but rising power intensity is making energy-efficient scaling and advanced packaging more critical to the next leg of growth. Applied Materials (covered by UBS analyst Tim Arcuri) is positioned as an enabler of this transition through materials engineering, GAA/backside power, thermal management, hybrid bonding, CPO, and system-level packaging. The takeaway is that AI infrastructure capex should increasingly flow into semiconductor equipment and packaging technologies that solve power, bandwidth, and integration bottlenecks.

- **Industry expansion has accelerated.** The industry took 50 years to reach US \$500bn and only 4-5 years to reach US\$1trn. The cycle should continue as new applications and technologies drive incremental demand. Agentic AI is becoming a rack- and data-center-level problem.
- **Optimization dilemma: node scaling versus power delivery.** Performance cost has improved 3x through technology scaling, advanced packaging, and new architectures, while power consumption has also increased 3x. The speaker noted that the cost per TFLOPS per GPU fell from US\$180 to US\$40 between 2016 and 2025, while the power cost per GPU hour increased from US\$0.05 to US\$0.35 due to higher chip power.
- **Applied is focused on improving energy-efficient performance from first principles.** Semiconductor technology is targeting lower capacitance, voltage reduction through gate-all-around, lower-resistance back-end-of-line interconnects for better conductivity, and improved thermal management in 3D integrated circuits. Thermal management accounts for roughly 40% of data-center operating cost. Key architecture innovations include GAA transistors, backside power delivery, HBM, vertical transistors, 3D DRAM, and advanced

packaging.

- **Advanced packaging complexity is ramping rapidly.** Since the 2000s, and through future innovations, package size has scaled up 9x, silicon area 600x, die count 400x, and transistor density 1,000x. SiP will need to incorporate hybrid bonding, CPO, embedded passives/actives, glass substrates, through-glass vias, and bridge dies. Hybrid bonding is advancing from microbumps at 1,000 I/O per mm² to hybrid bonding at 10,000 to 1mn I/O per mm² at less than 10% of the power, or below 0.05pJ per bit. Next-generation packaging will incorporate HBM hybrid bonding, 3.5D IC chiplet stacking, and co-packaged optics.
- **Technology innovation cycles are speeding up.** Historically, major industry innovations such as FinFET took roughly 15 years to develop. Applied's EPIC R&D center, opening in 2026, brings the ecosystem together with the goal of compressing R&D timelines by 2x, focusing on materials and systems to pull in foundation research, equipment development, and module development.

ARM: From CPUs to AI Compute Platforms - Scaling Open Infrastructure with ARM

Speaker: Michael Wong, President of ARM Taiwan and SE Asia

Key message: Agentic AI is likely to expand the CPU opportunity as AI systems shift from accelerator-centric token generation toward broader orchestration, memory, networking, and system-management workloads. Arm (covered by UBS analyst Tim Arcuri) is positioning for this transition through Neoverse momentum, OCP engagement, and its Total Design ecosystem, which could help lower adoption friction as SoC complexity and design costs rise. AI infrastructure growth may increasingly benefit Arm-based server CPUs and the surrounding ecosystem, not just GPUs and accelerators.

- **Arm has been involved in OCP since its early days.** Arm began investing in infrastructure around OCP's founding in 2011 and joined OCP as a Platinum Member in 2018 when it launched Neoverse. Neoverse began deployment at CSPs in 2022 and has now shipped more than 1.5bn cores into data centers. Arm is also now on the OCP board, helping support OCP's 650+ members. The SoC design cycle is typically 24-36 months, with design cost increasing roughly 3x from US \$250mn at 7nm to US\$725mn at 2nm, and projected to exceed US\$1bn at 1nm.
- **Arm's Total Design program supports a broad ecosystem.** Arm Total Design spans foundries, design services and packaging, firmware and EDA, companion silicon, and OEMs. Partners include foundries such as IFS, Samsung, and TSMC; companion silicon providers such as Astera, Credo, Eliyan, Nokia, Rebellions, and SiPearl; and design service and packaging partners including ADTechnology, Alchip, Alcro, AMI, ASE, Asicland, Broadcom, Cadence, Faraday, GUC, HCLTech, Marvell, MediaTek, Nokia, Novatek, Realtek, SemiFive, Socionext, and Via Next.
- **Agentic workloads are increasingly taxing CPU architecture.** Arm projects that agents could scale from roughly 15 agents per human user today to tens of thousands per user, creating far more CPU orchestration demand as accelerators generate tokens and potentially swamp the system. Arm estimates this may require 3-5x more CPUs to balance agent flow.
- **Arm's AGI CPU processor is supported by a growing ecosystem.** The AGI processor uses TSMC 3nm, with 6GB of memory per core, 6TB of total chip capacity, and a 300W TDP. It has a 1U custom reference server based on the Open Rack V3 structure. The air-cooled version has 8,160 cores in a 36kW server, while the liquid-cooled version can scale to 45,696 cores at 200kW across 42 eight-node 1U servers. Arm's AGI CPU ecosystem includes partnerships with Meta, Oracle, OpenAI, F5, Cloudflare, Cerebras, Positron, Rebellions, SAP, SK Telecom, and Verda. Hardware and memory partners include ASRock, QCT, Lenovo, SMCI, AMI, Amkor, Micron, Samsung, and SK Hynix.

ASE: Advanced Packaging - Enabling AI at Scale (Sunny Lin, Christine Chen)

Speaker: Dr. CP Hung, VP of Corporate R&D

Key Message: AI-driven scaling is accelerating demand for advanced packaging, with fan-out packaging, CPO, and advanced power delivery shaping next-generation AI systems.

- **AI scaling is driving a step-change in advanced packaging requirements.** The industry is moving from scale-up, to scale-out, and ultimately scale-across architectures as requirements for performance, memory, area, power and thermal, are all increasing by roughly 2-10x.
- **FoCoS, FoCoS Bridge, and panel-level packaging are enabling the next generation of large AI accelerators.** ASE highlighted FoCoS (fan-out chip-on-substrate) and FoCoS Bridge as key technologies for heterogeneous integration, combining high-density bridge interconnects with redistribution layers (RDL). Compared with traditional flip-chip packaging, interconnect density increases by approximately 53x with FoCoS and 212x with FoCoS Bridge. The company also showed how panel-level packaging improves manufacturing economics and output, increasing the number of interposers per substrate and enabling larger format packages required for future AI systems.
- **Silicon photonics and co-packaged optics emerging as next generation system integration.** Dr. CP Hung highlighted co-packaged optics (CPO) and optical engines as key enablers for bringing optical connectivity closer to the processor. Currently 3D FOPOP can integrate EICs and PICs together to achieve edge coupling without TSVs. However, to support 200 Gbps per lane transmission in the future, TSVs will be required. ASE indicated that moving optics closer to compute engines can provide up to 32x higher bandwidth while reducing energy loss by approximately 6x, supporting future 200G-per-lane networking requirements and addressing signal integrity, thermal, and coupling challenges.
- **Vertical Power delivery a key upgrade to improve power conversion efficiency to the accelerator.** Dr. CP Hung stated that moving power delivery from a lateral to vertical architecture can reduce power loss from 10% to approximately 2% and save more than 30% of area in the X and Y dimensions although height may increase on the Z-axis. ASE is evaluating multiple approaches, such as embedding IVR chips and inductors into the substrate or RDL, to further shorten power delivery paths. The roadmap also highlighted evolution from multi-stage conversions (800V-54V-12V-1V) toward more streamlined power delivery architectures, supported by technologies including embedded power components, silicon substrates, silicon bridges, power SiP modules, and advanced package-level voltage regulation.

Broadcom: Open Ethernet Fabrics for AI Training and Inference / Driving Data Center throughput through 3.5D heterogeneous scale-in

Speaker: Mohan Kalkunte, VP of Architecture & Technology - Core Switching Group

Key Message: Broadcom (AVGO; Covered by UBS Timothy Arcuri) discussed the network configuration for scale-up in rack, scale-out in spine leaf across racks, scale across to connect data centers each connected by Ethernet. The company strongly believes the 3.5D integration is here and the 3D stacking will be the future of XPU's and can maximize the interconnect density.

- **AI system scaling is moving in three directions: scale-up, scale-out/across, and scale-in.** Scale-up connects more accelerators into a tightly coupled domain, scale-out/across links very large clusters across racks and rows, and scale-in packs more compute, memory, and I/O into a single package. Scale-up domains can move from roughly 72 accelerators to 144+ and beyond, while scale-out/across clusters can scale toward 10k-100k+ accelerators. Scale-in is equally important because the package is becoming the new system boundary: a processor package that is around 2,000mm² today could roughly double toward 4,000mm² over the next few years as accelerators integrate larger compute tiles, more HBM, higher-speed SerDes, and more on-package networking.
- **Ethernet is being adapted for AI scale-up through ESUN and related scale-up Ethernet work.** ESUN, or Ethernet for Scale-Up Networking, is an OCP effort backed by AMD, Arista, Arm, Broadcom, Cisco, HPE, Marvell, Meta, Microsoft, NVIDIA, OpenAI, Oracle, and others. ESUN v1 focuses on making Ethernet behave more like a deterministic scale-up fabric by improving small-message efficiency, lossless delivery, multi-hop congestion control, traffic differentiation, and link-level

reliability. Key technical elements include a compact ESUN header that reduces standard IP/UDP overhead for small packets, support for PFC and credit-based flow control, and link-level retry to reduce packet-loss exposure and improve accelerator goodput.

- **Optical Compute Interconnect is emerging as an open physical layer for multi-rack AI scale-up.** The OCI MSA was formed in March 2026 by AMD, Broadcom, Meta, Microsoft, NVIDIA, and OpenAI to standardize optical scale-up links and reduce reliance on proprietary copper interconnects. OCI targets a silicon-centric optical PHY that can support pluggable, on-board, near-package, and co-packaged implementations. Technically, OCI v1 uses NRZ signaling with wavelength-division multiplexing, bidirectional traffic over fiber, external laser sources, and optical-engine management/diagnostics so accelerators and scale-up switches can connect across racks with higher reach, lower power, and broader multi-vendor
- **Open cluster designs are moving AI infrastructure toward modular, mix-and-match systems.** Hyperscalers and infrastructure providers want hardware and software building blocks that can be combined across vendors rather than locked into one proprietary rack architecture. First-generation open cluster design specifications were introduced in January 2026, with second-generation work in progress across compute, networking, cabling, rack, management, and serviceability requirements. Participating companies include Ampere, Broadcom, Celestica, Dell, RS, Siemens, Toshiba, and others, with the goal of creating repeatable cluster blueprints for large-scale AI deployments.
- **CPO is becoming a practical path to lower-power, higher-reliability AI networking.** Broadcom has integrated optics into Tomahawk 6 Davissan, a 102.4Tbps Ethernet switch with co-packaged optics. By placing optical engines next to the switch ASIC, CPO shortens the electrical path, reduces trace loss and reflections, lowers the need for signal conditioning, and improves link stability versus front-panel pluggables. Broadcom has highlighted more than 50mn device hours with no link-flap errors, roughly 10x CPO reliability improvement versus pluggables, and more than 70% optical-interconnect power reduction, making CPO relevant for long-duration AI training jobs where even brief link events can idle thousands of accelerators.
- **Scale-in improves power, latency, and cost by moving more communication inside the package.** On-die and die-to-die connections can consume less than 0.2pJ/bit versus roughly 3-5pJ/bit for off-chip links, while die-to-die latency can be around 5ns versus roughly 30ns for chip-to-chip PHY links. This matters because AI accelerators are increasingly limited by moving data between compute tiles, HBM, cache, networking chiplets, and I/O chiplets rather than by raw compute alone. Higher-density SoCs and advanced packages can amortize fixed rack and platform cost over more compute, reduce board-level interconnect power, and make the package behave more like a single integrated accelerator.
- **Future XPUs will scale compute by combining larger packages, faster memory, higher network bandwidth, and 3D integration.** Current XPUs can already exceed 5,000 TOPS on 5nm/3nm-class silicon, with roughly 10Tbps of network bandwidth, 9TB/s of memory bandwidth, a 3,000mm² interposer, and around 2,000mm² of active silicon. Over the next two years, next-generation XPUs are expected to move toward 2nm/A14-class nodes, 25Tbps+ network bandwidth, 20TB/s+ memory bandwidth, 10,000+ TOPS, 6,000mm²+ silicon interposers, and 3,500mm²+ of 2D silicon, potentially exceeding 7,000mm² with 3D stacking. The key technical shift is that compute, HBM, cache, networking, and I/O will increasingly be co-designed as one package-level system.
- **3.5D XPU architectures use hybrid bonding and face-to-face stacking to raise interconnect density.** A typical 3.5D XPU can combine a base die that contains cache and network-on-chip functions, top compute dies, I/O chiplets for 800G/1.6T Ethernet logic, die-to-die PHYs, HBM stacks, and a large interposer on a multi-core substrate. First-generation face-to-back stacking connected the top silicon layer to the backside of another die through TSVs, limiting I/O density to available via counts and adding parasitic overhead. Second-generation face-to-face hybrid copper bonding can increase signal density by roughly 10x, reduce parasitics by moving the bonding interface closer to the active circuitry, and allow stacked dies to behave more like one piece of silicon. This makes 3D stacking a key

path to maximize bandwidth density between compute, cache, memory, and I/O.

- **3.5D/SolC integration is powerful but introduces yield, thermal, power, and design complexity.** Hybrid bonding can deliver very low-energy interconnects, but stacking multiple chiplets means yield loss compounds across 4-16 dies unless known-good-die testing, redundancy, and repair schemes are robust. Logic teams must also manage timing closure across chiplets built on different nodes, clock-domain crossings, and physical-design constraints created by dense die-to-die interfaces. Power delivery becomes harder because the same or higher current must be delivered through less area, while thermal dissipation becomes more difficult when active logic is stacked vertically. These challenges make power integrity, heat spreading, stress/warpage control, and package-level co-design essential for production readiness.
- **3.5D SolC is moving from concept toward readiness as power and latency benefits become compelling.** The company believes 3.5D integration is now practical because hybrid-bonded die-to-die links can reduce interconnect energy to below 0.1pJ/bit and latency to below 1ns, while reducing interposer and substrate footprint. Smaller interposers can lower cost and improve mechanical stress and warpage, although thermal, yield, and power-delivery challenges still need careful management. If executed well, 3.5D/SolC gives XPU vendors a path to keep scaling compute, memory bandwidth, and I/O density even as monolithic die scaling becomes more expensive and reticle-limited.

Google: The Fungible and sustainable Data Center - A Blueprint for the AI Era

Speaker: Gregory Moore, Google Platform Director

Key Message: Google (covered by UBS analyst Stephen Ju) discussed how AI adoption is pushing infrastructure requirements from compute alone into full-stack optimization across chips, networking, power, cooling, and energy storage. Its work on TPUs/GPUs, $\pm 400\text{VDC}$ sidecars, SST-based 800VDC conversion, layered storage, liquid cooling, and open standards shows how hyperscalers are redesigning the AI data center for higher rack density and better efficiency.

- **Google is seeing rapid token adoption across its API and applications.** Google said direct API usage by customers is now running at more than 22bn tokens per month, while Gemini is being embedded across Gmail, Search, Chrome, Photos, Maps, YouTube, Workspace, Drive, Pixel, and Waymo. At Google I/O, the company announced new AI features and infrastructure updates, including Antigravity, managed agents, Omni, Gemini 3.5 Flash, Gemini Flash, and TPU v8T/v8L. AI demand is broadening from standalone model access into productivity, consumer, cloud, and autonomous-driving workflows, which increases pressure on Google's compute, networking, power, and cooling stack.
- **Google is vertically integrating the AI infrastructure stack.** Google is deploying TPUs, internal and third-party CPUs, NVIDIA GPUs, first-party and third-party data centers, data-center networking, optical circuit switching, liquid cooling, and advanced power delivery. This gives Google more control over performance per watt, supply availability, and total cost of ownership across the full AI factory.
- **Google is building a layered energy-storage architecture for AI data centers.** Storage can sit at the chip/rack level through integrated capacitors, rack-level BBUs, and fast-response ride-through for transient GPU/TPU loads; at the row level through end-of-row or sidecar storage for peak shaving and bridging multiple racks; and at the facility level through large-scale BESS and microgrids for backup and grid support. It uses an energy storage hierarchy where capacitors respond fastest to microsecond-to-millisecond load steps, batteries provide longer ride-through, and BESS/microgrids help smooth facility-level demand and improve resilience.
- **Mount Diablo uses a $\pm 400\text{VDC}$ sidecar to enable much higher rack power.** Google, Meta, and Microsoft are collaborating through OCP on a sidecar power-rack specification targeted for launch in 2026, with the goal of scaling compute racks toward 1MW by 2028. The concept moves power conversion and battery backup out of the IT rack, freeing more space for XPU while distributing higher-voltage DC to reduce current, copper, heat, and voltage drop. This is a bridge from

today's 48/54V rack power toward higher-voltage DC data centers.

- **SST interface standardization is aimed at direct medium-voltage AC to 800VDC conversion.** Google, Microsoft, and NVIDIA released the OCP Solid-State Transformer v0.3 specification on July 30 as a starting point for standardization. The architecture defines input/output interfaces, power distribution, performance requirements, transient operation, mechanical dimensions, and physical interfaces. The technical goal is to bypass bulky legacy conversion stages, shrink the power footprint, improve grid-to-chip efficiency, and create common supplier requirements for LVDC-native AI data centers.
- **Google is standardizing liquid-cooling systems to support higher AI rack densities.** Google is now on its fifth-generation CDU with Project Deschutes, targeting a final v1.0 specification in 2Q26 and working with 10 suppliers to make the design available to the ecosystem. Deschutes is a high-capacity liquid-cooling distribution unit intended for dense AI/ML pods, while Project Brazos is a sidecar, closed-loop liquid-to-air cooling solution for older air-cooled facilities. Brazos supports about 60kW of thermal load per rack without requiring facility water, using modular cooling units, rack manifolds, leak detection, pressure protection, and remote monitoring.
- **Open standards help Google scale data-center deployments across a broader supplier ecosystem.** Google participates in at least 10 industry standards groups, including JEDEC, UCIe, OCP, RISE, Caliptra, CXL, CHIPS Alliance, NVM Express, UALink, and RISC-V, with more than 100 contributions to date. The strategic point is that open standards reduce vendor lock-in, improve shorten qualification cycles, and allow power, cooling, compute, memory, and interconnect suppliers to build to common AI-infrastructure requirements.

Microsoft: From Silicon to Systems Driving Innovation for Azure Compute and AI

Speaker: Gohar Waqar, Corporate VP of Cloud Hardware Infrastructure Management, Sushant Gupta VP of Hardware Engineering Planning and Architecture

Key Message: Microsoft is scaling Azure AI through vertical integration across silicon, systems, racks, data centers, power, cooling, security, and networking. Its custom silicon strategy, including Maia 200, Cobalt 200, and Azure Boost, is aimed at improving tokens per dollar, performance per watt, and fleet utilization while maintaining workload flexibility across CPUs, GPUs, and ASICs. AI infrastructure is becoming a full-stack systems race, with value shifting toward companies that can optimize compute, networking, power, cooling, and supply chain together.

- **Microsoft framed cloud infrastructure as a silicon-to-systems stack with a heavy APAC footprint.** The stack spans silicon, including logic, memory, and advanced packaging; building blocks, including power, thermal, and PCBAs; and systems, including servers and racks. APAC ecosystem spend has grown 8x over the past five years, and roughly 70% of full-stack components now come from APAC. Microsoft also has 12 live data-center regions in Asia, with new regions in Malaysia, Indonesia, India, and Taiwan.
- **Data-center investment is scaling rapidly to support AI demand.** Microsoft projected data-center capacity rising from 70GW in 2024 to 400GW by 2030, requiring more than US\$1trn of investment. AI model scale has increased roughly 10,000x from 2019 to 2025, while compute, storage, networking, and memory have each scaled only about 10x. The mismatch highlights why the bottleneck is shifting from any single chip to full-stack capacity, utilization, power, and networking efficiency.
- **Microsoft's strategy is vertical integration through system-level co-design.** The company is co-designing silicon, platforms, networking, racks, and data centers, with security and sustainability as foundational requirements. The technical goal is to optimize tokens per dollar, performance per watt, fleet utilization, and supply availability across the entire Azure infrastructure stack rather than relying only on merchant silicon.
- **Azure is giving customers optionality across multiple silicon choices.** Microsoft supports CPUs including Cobalt 200, AMD EPYC, and Intel Xeon, and accelerators including Maia 200, AMD Instinct, and NVIDIA Grace Blackwell. This

heterogeneous strategy lets Azure match different workloads to the right architecture: custom inference ASICs for token economics, GPUs for broad training/inference support, and CPUs for cloud-native and agentic orchestration workloads.

- **Microsoft's internal silicon targets inference, agentic CPU workloads, and infrastructure offload.** Maia 200 is designed for inference economics, with 30% better tokens per dollar versus leading GPUs, 1.4x performance-per-watt improvement through chip-to-model co-design, and the ability to power the latest OpenAI GPT models on Microsoft 365 Copilot. Cobalt 200 targets agentic and cloud-native workloads, with 33% lower agent-call latency, 50% performance improvement over Cobalt 100 for cloud-native workloads, and 23% higher throughput on agentic workloads. Azure Boost DPU offloads cloud storage and networking services from host CPUs, delivering 4x higher performance and 3x lower power while improving utilization, isolation, security, and hardware root of trust.
- **Microsoft is pairing hardware security with rack- and data-center-level systems innovation.** System-level work spans power, including high-temperature superconductors, solid-state transformers, Mt. Diablo racks, and high-temperature fuel cells, and cooling, including next-generation heat-exchange units and microfluidics. Hardware security capabilities include confidential computing, secure management controllers, integrated hardware security modules to protect keys, Caliptra open-source silicon root of trust, and PCI-SIG Azure Boost support for confidential-computing I/O acceleration.
- **Microsoft is pushing networking innovation across scale-up and scale-across AI fabrics.** The focus is on open standards and architectures for scale-up, including OCP, ESUN, and MRC, and data-center scale-across connectivity, including OCI-MSA, UALink, and the Ultra Ethernet Consortium. Better fabrics raise accelerator utilization, reduce stragglers and congestion, and lower training and inference cost by improving bandwidth, latency, reliability, and multi-vendor

NVIDIA: Scaling the AI Factory - Open Networking for the GigaScale Era

Speaker: Gilad Shainer, Senior VP of Networking

Key Message: NVIDIA (covered by UBS analyst Tim Arcuri) is positioning the AI data center as a full-stack "AI factory" built around open standards, modular MGX racks, 800VDC power, liquid cooling, NVLink scale-up, Spectrum-X scale-out, and CPO-based optical networking. The strategy extends NVIDIA's architecture beyond GPUs into rack-scale systems, power delivery, networking fabrics, optical interconnects, and semi-custom silicon through NVLink Fusion. AI infrastructure leadership is shifting toward companies that can optimize the entire data-center stack, not just individual accelerators.

- **NVIDIA participates with OCP on infrastructure, networking, and software standards.** NVIDIA is using open collaboration to co-design the AI factory from grid power to racks, transport protocols, management, and software. Its standards work spans 800VDC power, hardware security, open network install environments, open accelerator and NIC designs, liquid cooling, and hardware management. The broader technical goal is to make high-density AI racks deployable across a larger supplier base while improving serviceability, and time to market.
- **MGX provides an open, flexible rack architecture and partner ecosystem for pod-scale AI systems.** NVIDIA's MGX architecture supports modular compute, switch, and networking trays built around NVLink, Spectrum-X, direct-to-chip liquid cooling, and rack-level serviceability. The ecosystem includes more than 300 partners, single-wide rack dimensions, fanless and cable-free trays, and 45°C liquid-cooling support versus roughly 35°C on some alternative platforms. The technical value is that common mechanical, thermal, power, and management interfaces let customers mix GPU, CPU, DPU, NIC, switch, and storage building blocks across generations without redesigning the full rack.
- **800VDC is a priority for higher compute, networking, and power density.** NVIDIA is building ecosystem alignment around 800VDC data-center power so operators can start with hybrid deployments and later move toward native high-

voltage DC facilities. 800VDC supports higher compute density, scalable rack power, flexible power options, simpler electrical infrastructure, better power utilization, and improved resilience. Technically, higher-voltage DC lowers current for the same rack power, reducing copper, cable bulk, voltage drop, conversion losses, and heat versus 54V rack distribution or legacy AC-heavy architectures.

- **Gigascale AI factories combine scale-up, scale-out, and scale-across networking.** NVIDIA treats the data center as the computer, with the network defining how much useful compute the AI factory can deliver. Scale-up uses NVLink Switch for tightly coupled GPU domains; scale-out uses Spectrum-X Ethernet, BlueField DPUs, and AI-native storage/context-memory systems; and scale-across uses Spectrum-X switches and ContextX/ConnectX SuperNICs to connect multi-site or multi-cluster infrastructure. Sixth-generation NVLink delivers up to 3.6TB/s per GPU of all-to-all bandwidth for massive compute with low latency, while Spectrum-X supports AI Ethernet fabrics using 102T-class switches and 1.6T SuperNICs. Spectrum-X also supports multiple software ecosystems, including FBOSS, SONiC, Cumulus, and NX-OS.
- **Spectrum-X Photonics is moving co-packaged optics into production AI networking.** NVIDIA's CPO approach integrates silicon photonics close to the switch ASIC, using TSMC COUPE, micro-ring modulators, and high-power external lasers to shorten electrical paths and reduce optical/network power. NVIDIA highlights 4x fewer lasers, roughly 5x lower power, 10x higher mean time between repair, and stronger signal integrity versus pluggable optics. Spectrum-XGS for scale-across uses 800Gb/s Ethernet switch ports and ConnectX-9 SuperNICs, targeting roughly 1.9x multi-site performance by improving congestion control, resiliency, and bandwidth density across AI clusters.
- **NVLink Fusion opens NVIDIA's rack-scale architecture to semi-custom accelerators and CPUs.** NVLink Fusion lets customer XPU connect into NVIDIA's NVLink scale-up fabric, while NVLink-C2C connects CPUs such as NVIDIA Vera or partner CPUs into the platform. Fusion allows third-party silicon to plug into NVIDIA networking and MGX rack architecture, including compute trays, switch trays, spine connectivity, and multi-GPU rack integration. The ecosystem includes optical interconnect partners such as Ayar Labs, Lightmatter, and Marvell; technology partners such as Cadence and Synopsys; custom-silicon partners such as Alchip, Astera, GUC, Marvell, MediaTek, and Samsung; and CPU partners such as Arm, Fujitsu, Intel, and SiFive.

TSMC: Advancing 3DIC Technologies to Propel AI Innovations (Sunny Lin, Christine Chen)

Speaker: Dr. Jun He, VP of Advanced Packaging Technology and Service

Key Message: TSMC (2330.TW, covered by UBS Sunny Lin) continues to ramp advanced packaging with excellent yields, while the industry faces challenges in heterogeneous integration, reliability qualification, and component supply constraints.

- **The AI-driven semiconductor opportunity is broadening beyond traditional HPC, supporting a >US\$1.5tn industry forecast by 2030.** TSMC estimates the semiconductor market will exceed US\$1.5tn by 2030, with HPC/AI representing approximately 55% of the opportunity, while the remaining ~45% will come from a broad range of applications including, but not limited to, smartphones, automotive and IoT. Dr. Jun He described this trend as "intelligence in every device."
- **CoWoS and SoIC are ramping fast with excellent yields.** Dr. Jun He said TSMC currently has 10 advanced packaging plants, and capacity continues to ramp quickly, with CoWoS and SoIC capacity expected to grow at an over 80% and 90% CAGR, respectively, from 2022-2027F. Reiterating what was presented at the [symposium](#), the R&D team is working toward CoWoS solution that could support >14 reticle-size package with 25 HBMs by 2029. Meanwhile, SoIC with 6um pitch has already been in high-volume manufacturing since last year. SoIC with hybrid bonding offers more than 50x interconnect density and 5x greater power efficiency.
- **Heterogeneous integration is increasing manufacturing complexity as package size and integration density continue to grow.** Dr. Jun He highlighted that increasing integration density and complexity lead to greater

mismatch gradients, while larger form factors and shrinking process margins intensify thermal and mechanical interactions.

- **Component-only qualification is no longer adequate as system-level reliability becomes increasingly important for AI applications.** Dr. Jun He emphasized that component-only qualification is no longer sufficient for AI applications, and that gaps between component- and system-level reliability must be addressed through upfront risk mitigation.
- **ABF substrate and memory supply constraints will remain an industry challenge.** Dr. Jun He highlighted that shortages of memory and ABF substrates are expected continue over the next couple of years. As the supply chain increasingly takes on the risk of multi-sourcing, Dr. Jun He encouraged industry participants to work closely with their sourcing teams to address issues in thermal, power and SMT (surface-mount technology).

Power electronics track (Randy Abrams, Irene Chang)

We attended the power electronics track at the 2026 OCP APAC Summit, where Microsoft, Google, NVIDIA, Delta, Hon Hai, Lite-On, Renesas, and Vertiv discussed the theme that AI rack power is outgrowing legacy AC and low-voltage distribution and will drive an upgrade to the power electronics architecture over the next few years. The panel on 800VDC framed the transition as a necessary step for higher-density AI factories, with panelists from NVIDIA, Delta, Google, and Microsoft seeing initial 400+/- and 800VDC implementations in the coming 1-2 years with a side power rack delivering DC power to the IT rack and larger upgrade of the grey space in 2028-30 with a solid state transformer delivering 800V DC throughout the data hall and to the server boards.

Among the panelists, Delta focused on grid-to-chip efficiency and SSTs, Hon Hai on liquid-native AI pods, Lite-On on server PSU-to-rack-power migration, Microsoft on Mt. Diablo validation, Renesas on high-density GaN/DC-DC conversion, and Vertiv on sidecar and power-center architectures. The common message was that power is no longer just facilities infrastructure—it is now a core variable for accelerator utilization, rack density, build speed, and data-center TCO.

The main architectural shift is from distributed low-voltage conversion toward 400V/800VDC sidecars, power racks, and eventually SST-enabled high-voltage DC facilities. Higher voltage reduces current, copper, cable bulk, I²R losses, heat, and rack-space consumed by power shelves and backup systems. Near term, adoption is likely to start with sidecars and rack-level DC/DC conversion, because brownfield data centers can retrofit this model without a full rebuild. Longer term, native 800VDC servers, centralized power centers, SSTs, BESS, CBUs/BBUs, and layered capacitor/supercapacitor storage should enable multi-hundred-kilowatt to megawatt-class racks with better ride-through and fewer conversion stages.

800V DC and Beyond: Building the Native DC Data Center (Panel with Microsoft, Google, NVIDIA and Delta)

Speaker: Microsoft: Kelvin Huang, Sr. Director, NVIDIA: Roy Luo, Infrastructure Electrical Lead, Google: Walter Lai, Hardware Manager, Delta: Peter Barbosa, VP of R&D

Key message from the 800V DC panel. 800VDC power distribution is emerging as a key enabler for higher-density AI data centers, with Microsoft, Google, NVIDIA, and Delta highlighting efficiency gains, reduced copper and cabling, fewer power-conversion stages, and more usable rack space for compute. The panel broadly agreed that near-term adoption will start with sidecar and DC rack architectures, while longer-term progress toward a full 800V DC data center architecture depends on ecosystem readiness around OCP standards, solid-state transformers, busways, battery backup, safety certification, and DC arcing protection.

NVIDIA appears ready to deploy 800VDC beginning with Vera Rubin-era rack designs in the coming year, with Microsoft, Google, and Delta agreeing on that initial adoption with side power racks and broader adoption throughout the grey space of the data

center with a solid state transistor delivering 800V DC having supply-chain maturation and standardization readiness toward 2028 and potentially wider deployment before 2030.

Why 800V DC?

- **NVIDIA.** The reasons to switch from 480V AC to 800V DC are: 1) efficiency gains, as AC is subject to the skin effect, using only the surface area to conduct current, making it less efficient; 2) 800V DC reduces conductor requirements and uses less copper, as the higher voltage can carry more MW at the same current; and 3) DC also avoids inefficient conversions in Uninterruptible Power Systems (UPS) between AC and DC. To power a 330kW rack, AC power would require eight 100A whip cables, plus more fiber optics and cables in a narrow space, whereas 800V conductors deliver power more effectively. 800V offers efficiency gains, high-density power deployments, and reshapes AI factory design. AC has limitations in delivering more power to the rack, which constrains compute density. DC has advantages in reducing the need for copper to conduct power and also frees up space in the IT rack. The certification path is also easier for IT equipment.
- **Delta.** 800V DC is a good choice for scalability and can significantly simplify the power chain by saving 3–5% efficiency when moving from AC to DC distribution. Replacing an AC-DC shelf with a DC-DC shelf can also save significant space in the rack that can be used for additional compute capacity.
- **Google.** The most important factor for 800V DC is balancing the data center facility's power architecture to support the technology generation. 800V, 400V +/- solutions, and 48V power delivery from 12V are core challenges in finding an appropriate balance at this point. From an infrastructure perspective, high voltage must go through multiple layers of equipment, which can be made more efficient with fewer conversions using DC power, while also improving build speed and enabling smaller, faster data centers. The impact extends beyond the rack level and will affect AI factory design.
- **Microsoft.** Current = power / voltage. A higher voltage can drive more power. From a rack real estate perspective, the scale-up fabric is bounded by the space in a rack unit, and every unit spent on a power shelf and battery backup takes away from compute capacity. Rack space is the key driver of moving power out of the rack. A conversion stage with fewer components can also drive higher efficiency and easier maintenance.

Evolution path for 800V DC

- **Microsoft.** Mt. Diablo is a standard spec with key players on board. For 800V DC, the ecosystem needs to be ready. SST, busway, and battery backup development also need to come together to enable the full transition. The safety standards need to be extended from 600V to 800V, and SST may need to go through the outline of investigation process, which could take a few years to establish the safety standard.
- **Google.** Over the past two years, Google's focus has been Mt Diablo with Microsoft and Meta, and it has driven the collaboration with partners to mass produce the sidecar in the data center for TPUs and GPUs. It has co-authored a white paper with NVIDIA and Microsoft on the SST specification as part of a joint effort aligned with industry partners. Safety remains on the critical path, so it is working with partners, particularly to address DC arcing.
- **NVIDIA.** The ecosystem spanning from cloud partners and equipment OEMs are working toward the same goal, and standards such as OCP are helping OEMs like Delta create and mass produce the product. Steps will be from sidecar to power center/power rack, and then to SST.

Readiness for 800V

- **NVIDIA.** NVIDIA is ready. The VR rack will have AC power supply and DC power supply options and can have a sidecar with two Vera Rubin racks on the side. This will be the first step toward evolution to an architecture with a solid-state transformer for high-density power and efficiency. NVIDIA is starting at the rack level with Vera Rubin. Next, it is looking at the DSS reference architecture (NVIDIA AI factory blueprint), and the upcoming VR release will have AC and DC solutions

to use as a reference for how to deploy power sidecars and power centers. 800V is being implemented now, ahead of the SST.

- **Delta.** The power side car today is an initial step and can help gain experience in DC distribution. The next stage is developing LV DC expertise to use a TRU and, when mature, can provide a path to adopting the SST. In terms of timeline, Delta has been working on the technology since 2022, and proofs of concept are already underway with customers in their local facilities and university labs. It believes that, in the next year or so, it can provide more advanced products for SST. Deployment may be before 2030, and the company is working hard to get it through certifications on medium voltage. Side car architectures will be important, with LVDC and, hopefully, SST becoming the preferable technologies for data centers down the line.
- **Google.** Over the next few years, the GPU and TPU programs will adopt 400+/-DC solutions. The flexibility on the sidecar can still have a MW rack in the data center without rebuilding the data center. It will still adopt sidecars over the next few years. LVDC is Google's long-term vision to eliminate the sidecar in the white zone to have more space for the IT rack. Internally, it is aligned with the data center and IT teams to make a smooth, production-ready decision.
- **Microsoft.** Key to readiness in 2028 is supply chain readiness and SSTs moving from prototypes to production. Some are working on 13.8kV, but the key enablers will be 25V and 34.5V to support mass deployment, so the voltage gap is another issue. High-voltage DC certification must go beyond component-level certification and extend to system-level certification, which still requires significant work. Regarding timing, Microsoft's view is also aligned with Delta's, with PoC, prototypes, and standards expected to have some setups ready for SST to support 800V DC by 2028.

Technology hurdles to ensure 800V safety

- **Delta.** The shelves have significantly higher power density, which increases EMI, extends certification timelines, and requires additional protections to detect faults. Arc flash is always present and must also be addressed.
- **NVIDIA.** The safety voltage is 50 V, and anything above that is unsafe; however, the data center is already handling higher voltages. This requires safety practices, certifications, equipment (circuit breakers, DC relays), and operating best practices.

Delta: Power Distribution and its Interface with Rack Power Systems

Speaker: Peter Barbosa, Rudy Wang, Director of R&D, Aurelio Rodriguez Echevarria, Power Architect

Key Message: Delta Electronics (2308.TW; Covered by UBS Randy Abrams) discussed how AI data-center power is moving from traditional low-voltage AC distribution toward higher-voltage DC architectures as rack power rises from hundreds of kilowatts toward megawatt-class levels. 800VDC, solid-state transformers, local energy storage, and hierarchical ride-through are needed to reduce copper, conversion losses, heat, and transient instability from fast GPU load swings. AI infrastructure spend with these upgrades is benefiting power electronics, DC distribution, energy storage, connectors/protection, cooling, and OCP-standardized power systems.

- **Challenges powering the AI data center span the full path from grid to chip.** The key issues include utility/grid supply, AC/DC power architecture, distribution to the IT cabinet, backup and ride-through for fast load swings, power quality and power-factor correction, protection/safety, and cooling. As AI racks move from hundreds of kilowatts toward megawatt-class loads, power delivery becomes a system-level constraint rather than a facilities afterthought.
- **AC power architecture is evolving from multiple low-voltage conversion stages toward higher-voltage DC distribution.** Today's grid-to-chip path typically converts 400/480VAC through UPS/PDU and power-shelf stages to roughly 50-54VDC, then steps down through 12V/6V rails and finally through VRMs to sub-1V for the XPU, with cumulative losses and about 87.6% efficiency in the referenced architecture. The upgrade path adds an in-row power rack with optional BBU/PCS, while the longer-term architecture moves toward a solid-state transformer outside or near the facility, energy storage replacing parts of the UPS/

PDU function, and an 800VDC bus feeding a DC power shelf that steps down to 50/54V or 12V before the final 0.65V-class chip supply. Higher voltage reduces current for the same power, which lowers copper, heat, voltage drop, and distribution loss.

- **DC power architecture in 800V IT racks can follow either a 54V busbar path or a native 800V rack path.** In the first option, AC power is converted to 54V and distributed on a vertical busbar with fuse and hot-swap protection, then stepped down to 12V and finally through VRMs to sub-1V for the GPU/XPU. In the second option, a native 800V vertical busbar feeds the server, and in-server conversion steps 800V down to 48V, 12V, or 6V before the VRM supplies the XPU at sub-1V. Both 54V and 800V busbar approaches can work around the 200kW to 500-700kW range, but above 500-700kW the 800V-to-rack approach becomes more likely to dominate because current at 54V becomes difficult to manage. In this implementation, a high-voltage bus converter steps 800V down to 48V, followed by an unregulated bus converter and voltage regulator supplying the XPU.
- **DC distribution must support both continuous operation and fast transient operation.** In continuous operation, the power-conversion chain supplies steady-state load while maintaining voltage regulation and power quality. In transient operation, the system must ride through rapid GPU load steps, source transfers, and protection events while different sources stabilize current over tens of milliseconds. A second transient case can occur when a rack or downstream fault farther away from the source causes a protection trip, surge, or dropout lasting hundreds of milliseconds, requiring coordinated protection, energy storage, and voltage-hold-up design.
- **LVDC input power-supply architecture requires protection, filtering, hold-up, and safe hot-swap behavior.** Key components include the input connector, EMI input filter, hot-swap controller, inrush-current limiting, fuses or electronic protection, decoupling/hold-up capacitors, and the low-voltage DC/DC conversion stage. The input side requires a high-current, touch-safe connector with correct creepage/clearance, low contact resistance, arc mitigation, and serviceability because 54V and especially 800VDC distribution create different safety and protection requirements than traditional AC power.
- **Energy storage is required in DC distribution to meet grid-side requirements and smooth AI load swings.** Storage can use capacitors, supercapacitors, battery backup units, PCS systems, and BESS deployed at the rack, row, distribution, or facility level. Distribution-level storage can connect near the solid-state transformer to smooth power swings and support grid interaction, while rack- or row-level BBU and power-capacitor shelves can respond faster to GPU transients, ride-through events, and short dropouts closer to the load.
- **Hierarchical energy storage should match each power event to the right response time and energy capacity.** Capacitor trays can suppress voltage ripple and very fast microsecond-scale events, while supercapacitor trays can compensate GPU transients with millisecond response. Row-level storage can support 10ms to 1s cluster power balancing, distribution-level storage can provide seconds-to-minutes grid support and ride-through, and utility-scale BESS can provide minutes-to-hours energy management and backup. This hierarchy matters because no single storage technology optimally covers fast transient response, ride-through, and long-duration backup.
- **OCP's power-distribution workstream is aligning the ecosystem around common 800VDC requirements.** The workstream includes expertise across system architecture from Eaton, voltage levels and power quality from Siemens, protection and safety from ABB, main components from Schneider, control and operation from Vertiv, and joint power-delivery work from Delta and AMD. The goal is to standardize interfaces, safety, protection, power quality, and operations so hyperscalers and suppliers can scale 800VDC architectures without bespoke designs for every customer. Separately, Google noted rapid token adoption across its API and applications, which reinforces why AI infrastructure power density is becoming a more urgent platform-level issue.

Delta: Enabling Scalable AI Data Centers Through Solid-State Transformer Technology

Speaker: Peter Barbosa, Rudy Wang, Director of R&D

- **AI power demand is forcing a redesign of the data center electrical architecture.** Delta showed industry rack power growing from historical 15-50kW racks operating at 12V/54V to 140-200kW today, with roadmaps pointing to 600kW-1.2MW racks around 2028-2029 and potentially 1.5-2MW racks in the early 2030s. Existing UPS, PDU and distributed power architectures become increasingly inefficient and complex at these power levels, creating the need for a new electrical stack.
- **Solid-State Transformers (SSTs) transform the transformer from a passive component into an intelligent power node.** Unlike conventional transformers that only change voltage levels, SSTs incorporate power electronics to provide voltage regulation, fault protection, power-quality control, grid-forming functions, renewable integration and dynamic power management. As AI data centers become gigawatt-scale loads, power delivery itself is becoming a software-defined control problem rather than simply an electrical infrastructure problem.
- **The economic driver is replacing copper with semiconductors.** Delta argued that conventional transformers rely heavily on copper and large magnetic structures, while SSTs leverage advances in silicon-carbide power semiconductors and high-frequency magnetics. Operating at much higher frequencies dramatically reduces transformer size, weight and copper content, potentially creating a long-term cost curve more analogous to semiconductors than traditional electrical equipment.
- **SSTs simplify the power path and improve end-to-end efficiency.** Delta's architecture collapses multiple infrastructure blocks by integrating functionality traditionally provided by transformers, UPS systems, power racks and power conditioning equipment. Their example showed overall "grid-to-chip" efficiency improving from roughly 89.1% in today's architecture to about 92.1% using direct 800VDC distribution enabled by SSTs. While a few points may seem small, at AI campuses consuming hundreds of megawatts this translates into substantial reductions in power costs and cooling requirements.
- **High-voltage DC distribution becomes increasingly compelling at AI scale.** Moving from lower-voltage distribution toward 800VDC significantly reduces copper losses and cable requirements. Delta estimates approximately 60% lower distribution losses versus traditional low-voltage approaches, while enabling power delivery across much larger clusters. The transition is particularly relevant as future AI facilities scale from today's hundreds of megawatts toward multi-gigawatt campuses.
- **The key technical challenge is reliability, not basic feasibility.** SSTs target roughly 10kW/L power density and around 99.7% transformer efficiency, versus conventional transformers operating near 0.11kW/L and approximately 99% efficiency. Achieving this requires solving insulation, thermal management and fault-handling challenges under dramatically higher power densities. Delta's modular architecture includes hot-swap capability and N+1 redundancy to maintain system availability, and the company noted that a joint SST specification backed by major hyperscalers including Google, Microsoft and NVIDIA was published through OCP in June 2026, signaling growing industry alignment around the technology.

Hon Hai: AI data center infrastructure with high power and full liquid cooling solution

Speaker: Chris Pai, VP Engineering

Key Message: Hon Hai (2317.TW; Covered by UBS Randy Abrams) developing into data center full system solution provider alongside the data center evolution trend with AI training and inference are moving from the server level to the cluster level. AI data centers now require integrated infrastructure systems where compute utilization depends on power stability, liquid cooling, networking, and real-time operations. As rack density rises, the architecture is shifting toward higher-voltage DC distribution, power sidecars, direct-to-chip cooling, CDUs, dense telemetry, and workload-aware control.

- **AI workloads define the infrastructure requirements.** Training jobs require long duration, high accelerator utilization, and synchronized all-to-all communication, so the facility must be designed around uptime, latency, power

stability, and thermal headroom. Key implications include: 1) power—higher rack feeds, fault-current management, busway/whip planning, power quality, and battery/generator sizing; 2) cooling—direct-to-chip heat capture, CDU capacity, water chemistry, leak detection, and residual air cooling; 3) network—low-latency, high-radix switching with congestion control and telemetry; and 4) operations—real-time monitoring, workload-aware controls, spares strategy, commissioning, and acceptance testing.

- **Power architecture in the data center is evolving toward higher-voltage distribution and disaggregated power racks.** The roadmap is moving from 380VAC and 480VAC toward $\pm 400\text{VDC}/800\text{VDC}$, standalone power sidecars, and high-voltage power delivered closer to the compute rack. The power chain runs from the grid through UPS/BESS, switchgear/busway, power shelves, and GPU trays, with design goals of safety, density, observability, and serviceability. Higher-voltage DC reduces current for the same power, lowering copper use, voltage drop, I²R losses, and heat, but it also increases the importance of protection coordination, grounding, arc mitigation, touch safety, and fault isolation.
- **Cooling is shifting from air to hybrid and liquid-native designs as rack density rises.** The liquid-cooling stack includes a server-side technology loop with cold plates, microchannel heat transfer, manifolds, quick disconnects, flow balancing, coolant, and leak detection; a CDU with pumps, filters, heat exchangers, reservoirs, sensors, and controls; and a facility loop using dry coolers, chillers, cooling towers, thermal storage, or heat reuse. The server-side loop must remain separated from the facility-water loop through a heat exchanger to protect IT equipment from water-quality issues, pressure events, and contamination. Facility-side design considerations include CDU location, pipe and valve layout, air/liquid integration for residual heat, coolant chemistry, pressure and flow budgets, warmer-water economization, and heat-rejection strategy by rack module.
- **A liquid-native AI pod integrates compute, power, cooling, and control as one operating system.** The pod has four planes: 1) compute plane—accelerator racks, storage, and CPU infrastructure; 2) power plane—medium-voltage input, UPS/BESS, busway, rack power shelves, and monitoring; 3) control plane—DCIM, BMS, telemetry, workload scheduler, and automation; and 4) cooling plane—facility water, CDUs, manifolds, cold plates, and leak detection. Liquid-native design lowers risk by planning for leaks, condensation, pump/CDU failover, power events, water quality, maintenance procedures, and human-error controls. Effective management requires dense sensors, a shared data platform linking BMS/DCIM/telemetry/workload scheduling, and a digital twin to simulate cooling scenarios and automate safe control actions.

Lite-On: Powering AI Infrastructure from power shelf to rack level power architecture

Speaker: Rafael Lee, Business Director

Key Message: Power demand is rising quickly as GPU power and rack density move toward multi-hundred-kilowatt and eventually megawatt-class levels. Lite-On (2301.TW, Not covered) showed how data-center power is shifting from server PSUs to rack-level power shelves, 800VDC power racks, BBUs, DC-DC bricks, and eventually SST-based high-voltage distribution. The broader implication is that AI infrastructure growth will increasingly depend on efficient rack-level power conversion, backup energy, and high-voltage DC architectures—not just compute performance.

- **AI is pushing data-center power demand sharply higher.** Energy consumption is expected to rise from 478TWh in 2024 to 1,590TWh by 2030, or roughly 5.5x Taiwan's power consumption. GPU power has also climbed from about 300W for A100 in 2020 to around 2kW for Rubin by 2026. Estimated new data-center capacity is stepping up from 12GW in 2024 to 16/24/32/38/43/50GW from 2025-30, with AI's share of data-center capacity rising from 38% to 60% of workloads.
- **Data-center system architecture is shifting from grid power to integrated energy systems.** Data centers need more grid capacity while adding wind, solar, turbines, hydrogen generators, and fuel cells. These sources require BESS to smooth intermittency, support ride-through, and stabilize fast AI load swings. Solid-state transformers can combine rectification and voltage transformation, taking medium-voltage AC from the grid or on-site sources and distributing it as

AC or high-voltage DC.

- **800V delivery to the rack still requires local DC-DC conversion before reaching the GPU.** Lite-On showed a migration from direct PSU to power shelf, then to an 800VDC power rack, and eventually to SST-based 800VDC distribution. In the first transition stage, a DC-DC rack is likely needed for the next 3-5 years to step high-voltage DC down for IT loads. Over time, rectification moves out to the facility “grey space,” reducing rack-level conversion complexity.
- **Lite-On’s roadmap scales from server PSUs to rack-level high-voltage power.** The roadmap starts with traditional 5.5kW server PSUs, moves to 33kW air-cooled power shelves for NVIDIA GB200, then to 110kW rack-level power shelves for GB300. As rack power rises, Lite-On adds 55kW liquid-cooled power modules and high-voltage rack architectures to support multi-hundred-kilowatt AI racks with better efficiency, lower distribution loss, and more available rack space.
- **Power distribution is migrating from traditional AC to 800V HVDC systems.** In today’s direct PSU model, AC is delivered to each server and converted locally to low-voltage DC. Power-shelf architectures centralize conversion in the rack, improving utilization and reducing PSU count. The next step is an 800VDC power rack, where high-voltage DC is distributed across the rack and local DC-DC stages feed accelerators. Longer term, SST-based distribution moves rectification and voltage transformation into the facility grey space, converting medium-voltage input directly to an 800VDC bus. This should reduce conversion stages, copper, cabling, heat, and distribution losses for multi-hundred-kilowatt and eventually megawatt-class AI racks.
- **Centralized 800V power racks can be deployed with or without BBUs.** BBU-enabled power racks integrate backup energy storage for ride-through, resiliency, and fast GPU load transients. Non-BBU versions focus on centralized power conversion and distribution where backup is handled elsewhere in the facility. Both options reduce reliance on individual server PSUs and simplify power management for large AI deployments.
- **High-voltage DC-DC bricks remain essential after the move to 800VDC.** Lite-On’s roadmap includes regulated and unregulated 800VDC-to-low-voltage bricks, with ratings rising from several kilowatts to more than 10kW per module. These bricks convert the 800V bus to intermediate rails such as 48V, 12V, or 6V before board-level voltage regulators supply GPUs and system electronics. This final conversion stage remains critical as accelerator power continues to rise.

Microsoft: Validating high power AI rack platforms with complex architecture: lessons from Mt. Diablo 400 System Qualification

Presenters: Venkata Nagesh Babu Doddapaneni Principle System Engineer and Jimmy Wang Senior System Engineer

Key Message: Mt. Diablo shifts AI rack power from the compute rack into a dedicated high-voltage DC sidecar, enabling up to 1MW racks with better density, efficiency, serviceability, and space for GPUs. The architecture requires rigorous L10/L11/L12 validation across PSUs, CBUs, PDUs, HVDC busbars, ride-through, fault isolation, grounding, and multi-rack safety. Megawatt-class AI racks will depend as much on a validated power architecture and protection design.

- **Mt. Diablo is a disaggregated power-rack architecture for high-density AI racks.** OCP Diablo 400 defines a $\pm 400\text{V}/800\text{V}$ -class HVDC sidecar power rack designed to support up to 1MW per IT rack, separate power conversion from the compute rack, and handle high-dynamic AI/GPU workloads. By moving power shelves, backup energy, and high-voltage distribution into a dedicated power rack, the architecture can free compute-rack space for more GPUs and switches while improving efficiency, density, copper use, and serviceability.
- **A structured validation framework is required at the component, rack, and system level.** Microsoft validates each building block at L10, including PSUs, CBUs, PDUs, and HVDC busbars; rack-level behavior at L11, including single-rack power delivery, ride-through, and fault isolation; and system integration at L12, including multi-rack validation across SKUs and workload profiles. Roughly 70% of validation is standardized component testing, 20% is design building-block validation, and 10% is application-use-case testing such as loading, redundancy options, grid capacity, handling, and management.

- **Validation must cover the full power chain and its new HVDC failure modes.** The building blocks include the AC PDU, PSU, CBU, DC PDU, HVDC busbars, and rack-SCM. Above the block level, validation must account for multiple SKUs, feed and PSU redundancy, HVDC power delivery, logic-level sideband signals, and management telemetry. New challenges include grounding schemes across multiple racks, complex DC fault behavior, high-slew-rate transients, dynamic responses across different IT-rack peak-power profiles, and coordinated rack-level features such as shutdown, isolation, and recovery.

Renesas: Advancing 3DIC Technologies to Propel AI Innovations

Speaker: Pietro Scalia Sr. Director of Power System Marketing and Marco Ruggeri, Senior Manager of Power System Architecture

Key Message: Renesas sees AI rack power moving from today's ~100kW class toward several hundred kilowatts, making 800VDC distribution and high-density DC-DC conversion necessary. Its roadmap spans 800V-to-48V/12V/6V conversion, modular 20kW platforms, and GaN-based designs with >98% efficiency to reduce losses and manage thermal load. AI infrastructure is increasingly depending on power semiconductors, wide-bandgap devices, and dense modular power architectures in addition to compute and storage silicon.

- **AI rack power is entering a step-function growth phase.** Renesas highlighted that accelerator and CPU power are scaling from today's ~100 kW-class racks toward several hundred kilowatts, with power content potentially increasing by up to 10x versus current GB200-era systems. The rise of multi-kW XPUs, chiplets, and increasing CPU:XPU power ratios is forcing a transition toward higher-voltage distribution architectures.
- **800V DC is emerging as the new backbone of AI datacenter power delivery.** Renesas is positioning itself across the full power chain, from grid-level conversion down to the processor, with solutions spanning 800V->48V, 800V->12V, and core-voltage regulation. This reflects the industry's shift away from traditional 48V-centric architectures as rack power densities become extreme.
- **GaN is becoming a critical enabler of next-generation power conversion.** Renesas showcased 6 kW converter reference designs for both 800V-to-48V and 800V-to-12V applications using combinations of high-voltage GaN, low-voltage GaN, and MOSFET technologies. The company reported >98% efficiency, highlighting why wide-bandgap devices are increasingly essential to control power losses in AI infrastructure.
- **Power conversion is moving toward highly modular, scalable architectures.** One example is Renesas's 20 kW 800V-to-6V DC-DC platform, which aggregates multiple 5 kW modules and parallel conversion stages. This modular approach allows power systems to scale with accelerating rack power requirements while maintaining efficiency and thermal performance.
- **The battleground is now power density, not just efficiency.** Renesas showed a roadmap of increasingly compact IBC and 48V converter designs, including new low-voltage GaN solutions capable of roughly doubling IBC power density. As AI systems become space-constrained and current levels surge, semiconductor integration and density improvements become as important as raw conversion efficiency.

Vertiv: 800VDC Power trends

Speaker: Henrik Nilen, Global Offering Director

Key Message: Vertiv (VRT.N; Covered by UBS Amit Mehrotra) sees AI power architecture moving from in-rack low-voltage power shelves toward sidecar and power-center models using 400V/800V DC to support multi-hundred-kilowatt and eventually megawatt-class racks. The transition will be staged, with brownfield sites likely adopting 800VDC sidecars first, while new-build facilities can move toward centralized HVDC power centers, BESS, SSTs, and on-site energy resources. The broader implication is that AI infrastructure scaling will depend heavily on high-voltage DC power conversion, ride-through, protection, and serviceable rack-level power architectures.

- **DC power evolution in the data center.** DC power in the data center has

evolved for more than 15 years from in-rack power shelves toward higher-voltage rack and pod architectures. Early OCP designs moved the UPS function closer to the IT load by replacing centralized UPS systems with BBUs near the rack, improving power density, backup locality, and rack efficiency. ORV1 started with 12V, 750W PSUs, a 4.2kW shelf, and an 85kW BBU rack with roughly 45 seconds of backup. ORV2 increased to 12V, 3.2kW PSUs, a 3.6kW BBU, and a 6.3kW PSU/BBU shelf with roughly 90 seconds of backup. ORV3 moved to 48V, 3.2kW PSUs and a 19.8kW PSU shelf, reducing current versus 12V while supporting higher rack density.

- Vertiv sees the AI power electronics evolving through five architecture steps.** Vertiv outlined the power progression upgrading the power per rack with 5 evolution paths: 1) 30-250kW rack with DC power shelf, 2) 30-140kW OCP standard to integrate the battery back-up with the power shelf and eliminate the UPS, 3) LVDC side car or centralized solution to feed 400V/800V into the power rack and suitable for 250-600kW with a power center integrating energy storage, capacitors (needed for AI storage) and energy storage to handle voltage for AI training with DC/DC converters in the rack to convert to 50V for the servers - the DC/DC converter is more compact vs. the PSU. The company sees a good roll out of side cars in the next year and a year later 4) native HVDC server to remove the DC-DC converter and use a PDU in the server rack and free up more space, and finally 5) 250kW-1MW rack which moves to a micro-grid on site with SST, renewables/fuel cells, BESS and get closer to 1,500V DC.
- Vertiv has a long history in higher-voltage DC power systems.** The company delivered its first 240VDC system in 2008 and its first China deployment in 2010, followed by first-generation 400V LVDC in 2012, second-generation 240V/400VDC in 2014, a third-generation 240V/400V 25kW rectifier in 2019, and APT 2.0 DCS with a phase-shift transformer in 2024. Vertiv then introduced its fourth-generation 240V/400VDC system in 2024 and is moving toward fifth-generation $\pm 400\text{VDC}/800\text{VDC}$ systems for 2025-26. This history matters because 800VDC is new for AI racks, but Vertiv already has experience with telecom, industrial, microgrid, and data-center DC power architectures.
- Mount Diablo is a $\pm 400\text{V}/800\text{V}$ -class OCP sidecar power solution for high-density AI racks.** The disaggregated sidecar architecture is being developed through OCP by Microsoft, Google, Meta, and the broader ecosystem. Different configurations can place power shelves in the rack, combine power shelves and energy storage in the rack, or move rectifiers and energy storage into a separate power rack. The system can scale toward 1MW and may include power conversion, BBU ride-through, capacitor-based transient support, DC PDUs, and high-voltage distribution. The key benefit is moving power out of the IT rack to free space for compute while improving power density, serviceability, and standardization.
- Each DC voltage range has trade-offs across efficiency, copper, safety, and standards maturity.** 0-400VDC is more mature and easier to protect, but it requires more current, copper, and conductor space at high rack power. 0-800VDC improves efficiency and reduces copper, voltage drop, and heat, but it requires stronger insulation, creepage/clearance, arc-flash mitigation, protection coordination, and updated standards. $\pm 400\text{VDC}$ combines some 400V and 800V benefits and can use a midpoint ground, but midpoint grounding adds protection and fault-management complexity. Some future IT loads may require true 800VDC input, including negative-ground architectures, especially as racks move toward native high-voltage servers.
- Hyperscale power architecture evolution path.** Hyperscalers should migrate their power through 4 steps: 1) In-rack 50VDC power shelves with BBU/CBU, 2) 50VDC power shelves, BBU, and CBU moves into a power sidecar with a copper busbar feeding the IT rack, 3) The sidecar upgrades to 800VDC with PSU, BBU, and CBU feeding the IT rack, where rack-level DC/DC conversion steps down for servers and 4) The side car feeds native 800VDC servers directly from the sidecar. Brownfield sites can continue using AC busways above the racks while DC is delivered by cables from sidecars. Efficiency progresses from roughly 94% for 50VDC, to 96.5% with in-rack BBU/CBU, to about 96% with a sidecar that enables much higher kW delivery, and back to roughly 96.5% when 800VDC feeds directly into the server.
- 800VDC sidecar and power-center roadmap separates brownfield**

deployment from LT greenfield architecture. DC Vertiv's 800VDC sidecar and power-center roadmap separates near-term brownfield deployment from longer-term greenfield architecture. The 800VDC sidecar is expected to support roughly 660kW and include hot-pluggable PSUs, BBUs, CBUs, and DC PDUs while still leveraging AC busways for brownfield sites. Vertiv also sees a liquid-cooled power version for white-space deployment arriving soon. A future power-center model could place multiple 1.6MW 800VDC power racks in the grey space and distribute power over a DC busway to IT loads, which is more likely in new-build facilities. The power-center approach can also pair with an energy center for BESS, renewables, fuel cells, or other on-site energy resources.

Photonics track (Sunny Lin, Christine Chen):

On Day 2 of the OCP APAC Summit, several leading photonics companies participated in a photonics track session. The presenting companies include Artilux, ASE, Ayar Labs, Broadcom, Coherent, Edgecore, Furukawa, GlobalFoundries, Lightmatter, Nvidia, NTT Devices, Seagate, and Synopsys.

Key message: CPO is emerging as a key solution to AI networking bottlenecks as copper reach, pluggable reliability, and chip-edge bandwidth constraints worsen at higher speeds. Moving optics closer to ASICs and accelerators could improve power efficiency, reliability, training uptime, and scale-up/scale-out flexibility across larger AI systems. For investors, the AI interconnect value chain may shift toward CPO, optical I/O, advanced packaging, and open standards that reduce power, cost, and vendor lock-in.

Below, we highlight our main takeaways from company commentaries:

Co-Packaged Optics (CPO) as the critical inflection point for power and cost efficiency

- **Broadcom:** "CPO is bringing optics closer to the ASIC, and that means it is much more power efficient, and it saves a lot of cost."
- **Ayar Labs:** "If I bring optical connectivity much closer in the scale-up fabric straight from XPU or accelerators, I can make a big jump up and to the right on this efficiency curve for networking. And then, if we also start to think about optically connected memory and storage appliances, I can get another big jump in efficiency. The whole point is to say that the integration of optical connectivity more deeply into these systems is going to deliver step changes in multiples of gains and efficiencies, not just incremental gains."

Reliability superiority over pluggable modules to improve training efficiency

- **Broadcom:** "Compared to pluggable modules, how does it work? If you see the mean time before failure (MTBF), CPO basically delivers more than 4x improvement. Because when you have a large data center, they plot diagrams of various pluggables and various configurations, and with that they are able to observe 90% training efficiency improvement as well."
- **Lightmatter:** "Pluggable optical transceivers are seen as the primary viability risk in data center networks today. But because they are so easily replaced, the problem really hasn't been solved."

Transition from electrical to optical scale-Up domains to bypass copper reach limits

- **Broadcom:** "The scale of domain is increasing from a single rack to multiple racks, and when that happens copper cannot offer a solution because it doesn't have that kind of reach."
- **Lightmatter:** "At 448 gig, our next increment is tiny – it's like 25 cm. Physics is pushing back."
- In our view Lightmatter is saying that traditional electrical signalling is running into physics limits at 448G SerDes speeds. At those data rates the signal can only travel a very short distance over high speed copper electrical cable —around 25 cm—before losses, noise, power, and signal-integrity

challenges become too severe. Lightmatter is using this to argue that optical interconnects are becoming necessary because they can move high-bandwidth data farther with better power efficiency and less signal degradation.

- **Ayar Labs:** Move the boundary of what the unit of adoption is from a full rack scale system over to a chassis connectivity. My unit of adoption and my kind of unit of simplicity becomes: I want to connect together a bunch of chassis."
- We view Ayar advocating to deploy optical I/O to link multiple racks together more simply and flexibly to make the rack unit smaller, easier to integrate and then connected into one high bandwidth system with co-packaged optics. We would note though scale-up domains like NVIDIA's NVLink used in its NVL72 is a high speed point to point network optimized for much lower latency, high bandwidth and fast GPU memory access. "Move the boundary of what the unit of adoption is from a full rack scale system over to a chassis connectivity. My unit of adoption and my kind of unit of simplicity becomes: I want to connect together a bunch of chassis."

Overcoming physical and mechanical packaging bottlenecks for high-speed connectivity

- **Lightmatter:** "As you scale up the size of the silicon, the area grows quadratically... the problem is with the bandwidth — the bandwidth comes out of the perimeter. The perimeter only scales linearly".
- In our view, Lightmatter is indicating you can scale compute by making silicon bigger, but you can't scale chip-edge bandwidth at the same rate—so interconnect becomes the bottleneck.
- **Ayar labs:** "We have electrical connections inside the rack, and that's where a lot of the constraints come from because we have the bandwidth-distance trade-off of copper connections. That drives significant complexity inside the rack as you put all the components very close together. So you're facing a power-per-rack challenge, and it makes it hard to scale up the size of the scale-up domain while facing physical limits of electrical connectivity".
- **ASE:** "The optical engines are very fragile. We have a few challenges here: package assembly yield, fiber connectivity, delicacy."

Collaborative industry standardization to prevent market Bifurcation and lock-in

- **Ayar Labs:** "Very excited about the OCI MSA which was founded by Meta, Microsoft, OpenAI, Nvidia, AMD and Broadcom... helping put some specifications and... common definitions."
- **Lightmatter:** " Our goal is that we don't want vendor lock-in; we want manageability, multiple technology support, and to be as open as possible but as performant as possible. This requires collaboration across IEEE, OIF, and all the MSAs. OCP is a great place to do it because it's neutral and open."

ASE: Inside the package - How CPO is powering the next wave of AI

Speaker: Nicole Tien. Senior Technical Program Manager

Key message: ASE (Covered by UBS analyst Sunny Lin) is positioning CPO as a natural extension of advanced packaging, moving optics, ASICs, switches, HBM, and power delivery into tighter system-level integration for AI servers. Manufacturing complexity rises materially around optical alignment, fiber coupling, yield, thermal stress, and multi-stage testing. AI networking bottlenecks could shift more value toward OSATs with end-to-end CPO, advanced packaging, and optical-test capabilities.

- **Packaging is expanding from chip assembly to system integration.** ASE illustrated packaging as the layer connecting XPUs, switches, optical engines, HBM, VRMs, and networking components into a unified AI platform. ASE positioned advanced packaging as the enabler of chiplet architectures, heterogeneous integration, power delivery, cooling structures, and co-packaged optics rather than simply attaching dies to substrates.
- **ASE has developed six major packaging platforms for AI.** The VIPack portfolio highlighted FoPoP, FoCoS, FoCoS Bridge, 2.5D/3D packaging, SoSiP, and CPO. Supporting technologies include multi-die integration, TSV-last processing,

advanced RDL carriers, ultra-low-loss interconnects, deep trench capacitors, and micro power-delivery structures that can be mixed and matched depending on AI system requirements.

- **HBM integration is driving larger package architectures.** The roadmap slide showed CoWoS-style packaging scaling from roughly one reticle in 2016 to 5.5 reticles today, with future packages approaching 10 reticles. This expansion enables larger compute complexes, additional HBM stacks, and significantly greater memory bandwidth while keeping manufacturing complexity manageable.
- **Silicon photonics is moving directly into the package.** ASE's roadmap progresses from optical engines located on PCBs, to optical engines mounted on substrates, and ultimately to optical I/O integrated beside ASICs. By shortening electrical transmission distances, this architecture reduces power per bit while supporting the higher data rates required for AI scale-out networking.
- **Optical packaging introduces new manufacturing challenges.** ASE highlighted that optical engines are mechanically fragile and sensitive to yield loss, thermal expansion, fiber attachment, and alignment errors. Slides focused on chip-on-wafer and wafer-to-wafer integration, double-sided testing, detachable fiber interfaces, and methods to maintain performance despite thermal and mechanical stress.
- **Fiber coupling remains a critical technical hurdle.** ASE is working on both edge-coupler and grating-coupler approaches. The company identified package warpage, coupling-angle tolerance, active versus passive alignment, fiber connectivity, and maintaining precise optical positioning throughout assembly as key issues that still need industry-wide solutions before large-scale CPO deployment.
- **ASE offers an end-to-end CPO manufacturing solution.** ASE's CPO capability spans the full optical packaging flow, including EIC die preparation, PIC TSV and middle-end-of-line processing, chip-on-wafer assembly for both electronics and optics, chip-on-wafer testing, wafer singulation, optical-engine functional testing, ASIC and optical-engine attach onto the substrate, lid and socket assembly, and final CPO functional testing and system-level testing. ASE can support customers from photonic die preparation through completed co-packaged optical modules.
- **CPO testing requires four dedicated insertion points.** ASE highlighted four major test stages: (1) EIC/PIC testing to validate individual electronic and photonic die, (2) chip-on-wafer testing after integration, (3) optical-engine unit testing to verify optical functionality and alignment, and (4) final CPO package testing after full system assembly. Key challenges include vibration sensitivity, scaling to multiple test sites, optical and electrical probe-card development, and precise fiber alignment throughout the test process.

Astera Lab: Optical Scale-Up Fabrics for Next Generation AI Infrastructure

Speaker: Jeffrey Kung, System Architect

Key Message: AI fabric is becoming a major bottleneck as accelerator clusters scale and copper becomes increasingly inefficient at 200G/400G per lane. Astera (covered by UBS analyst Natalia Winkler) is positioning NPO/CPO and high-radix optical switching as a way to improve reach, power, latency, and cluster utilization while integrating optics, switching, power management, and telemetry. AI networking spend is shifting toward optical connectivity, switch silicon, and system-level interconnect platforms that keep GPUs better utilized and reduce data-center power costs.

- **AI fabric constraints are becoming a key limiter to accelerator utilization.** AI model scale is expanding to tens of thousands of accelerators, while scale-up fabrics for inference are targeting thousands of accelerators. Agentic workloads add further pressure because they require low latency, all-to-all communication, high message rates, and persistent demand on the infrastructure fabric. As a result, bandwidth, latency, reach, and topology are increasingly determining whether expensive accelerators remain fully utilized.
- **Physics is catching up to copper at 200G and especially 400G per lane.** At 200Gbps/lane, long electrical channels already suffer from insertion loss,

reflections, crosstalk, and equalization burden, often requiring power-hungry DSPs or retimers. At 400Gbps/lane, those constraints become much harder, as the DSP can preserve signal integrity but adds power, latency, cost, heat, and board complexity. Short copper reach can constrain AI pods to a limited number of racks, particularly as flatter networks require longer lateral connections. Astera framed the “copper tax” as a roughly 650W power disadvantage versus CPO in a 10,000-accelerator pod, with LPO at 284W and NPO at 457W.

- **Optics addresses the reach, attenuation, and power limits that copper cannot solve economically.** Optical links maintain signal quality over longer distances with lower channel loss, enabling more flexible pod and cluster topologies. They also move heat away from dense switch and accelerator areas, reduce the need for retimers and DSPs in some architectures, and support higher bandwidth density as lane rates move from 200G toward 400G.
- **The optical migration path is moving from pluggables to NPO and ultimately CPO.** Pluggable optics are mature, field-replaceable, and available at high volume, but they keep a relatively long electrical path between the switch ASIC and the optical module, often requiring DSPs to condition the signal. NPO places the optical engine adjacent to the switch package, reducing the electrical interface from meters to millimeters and lowering power while preserving full optical reach across the fabric. CPO integrates optical engines inside the switch or accelerator package so fiber exits directly from the package, offering the lowest power per bit and highest bandwidth density, but with greater packaging, thermal, testing, and serviceability challenges. NPO is available now, while broader CPO availability is expected later in the decade.
- **High-radix optical switches enable flatter AI fabric topologies.** A higher-radix switch increases port count, allowing more accelerators, racks, or chassis to connect with fewer network stages. Fewer hops can reduce latency, improve bandwidth utilization, and simplify collective communication for training and inference. Optical I/O allows the pod to span multiple racks with better thermal distribution across the facility, retimer-free or retimer-light paths, and scalability toward tens of thousands of accelerators.
- **Astera’s integration approach combines switching, optics, power management, and telemetry.** Astera can integrate a high-port-count switch fabric ASIC with NPO, unified power management, and real-time telemetry. The platform can combine the switch ASIC, optical engine, management controller, and fabric software so that link tuning, diagnostics, firmware updates, and reliability monitoring are handled as one system. This matters because linear optics and NPO remove much of the module-side DSP intelligence, shifting more responsibility to the switch-side SerDes, system telemetry, and fleet-level management software.
- **The optical adoption timeline is moving from NPO now toward CPO and eventually fully optical AI clusters.** Astera sees NPO deployments happening now and becoming increasingly necessary as the industry moves toward 400Gbps/lane signaling. The expected roadmap is NPO adoption for 200G/lane in 2026-27, CPO switches with 400G/lane from 2026-29, on-die photonics over time, and potentially more fully optical AI clusters from 2030 onward.

Server track (Sunny Lin, Ryan Sun):

ASpeed: Enhanced server management and platform security architecture to strengthen its BMC moat (Sunny Lin, Ryan Sun)

- **New Root of Trust framework demonstrated at OCP.** Aspeed demonstrated the next generation Open Boot and Management Framework (OBMF) server management architecture and the Open Platform Root of Trust (OpenPRoT) platform security architecture at the 2026 OCP summit. The **OBMF** architecture focuses on standardized interfaces which enables the BMC and the host platform to exchange information more efficiently. This significantly reduces platform integration costs. As for the **OpenPRoT** security architecture, Aspeed showcases its AST1060 SoC with RoT firmware that has SPDm attestation capabilities which allows for tighter security integration across platforms which enables PRoT standardization. The OpenPRoT features will also be available on the upcoming AST1080 PRoT SoC. As for other progresses, we are positive on Aspeed’s [product roadmap](#) in preparation for the quantum computing era and its rollout of

AST2700 series BMC with [increased penetration in 2027](#).

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12-Month Rating	Definition	Coverage ¹	IB Services ²
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	40%	21%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 June 2026.

1:Percentage of companies under coverage globally within the 12-month rating category.

2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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Company Name	Reuters	12-month rating	Price	Price date
ASE Industrial ^{16,28a}	3711.TW	Buy	NT\$616.00	14 Aug 2026
ASPEED Technology Inc ^{28a}	5274.TWO	Buy	NT\$17,690	14 Aug 2026
Advanced Micro Devices Inc ^{16,7,28b,20}	AMD.O	Buy (CBE)	US\$514.39	14 Aug 2026
Alphabet Inc. ^{16,28a}	GOOG.O	Neutral	US\$343.54	14 Aug 2026
Applied Materials Inc ^{16,8,28a}	AMAT.O	Buy	US\$507.18	14 Aug 2026
Arm Holdings PLC ^{16,28a}	ARM.O	Buy	US\$279.44	14 Aug 2026
Astera Labs Inc ^{16,20}	ALAB.O	Neutral (CBE)	US\$321.61	14 Aug 2026
Broadcom Inc. ^{16,28a}	AVGO.O	Buy	US\$392.99	14 Aug 2026
Delta Electronics ^{4,28a,6c}	2308.TW	Buy	NT\$1,885	14 Aug 2026
Hon Hai Precision ^{4,5,28a,6c}	2317.TW	Buy	NT\$259.50	14 Aug 2026
MediaTek Inc. ^{28a}	2454.TW	Buy	NT\$4,210	14 Aug 2026
Microsoft Corp. ^{4,16,7,8,6a,28a,6b,6c}	MSFT.O	Buy	US\$495.40	14 Aug 2026
NVIDIA Corp ^{16,8,28a}	NVDA.O	Buy	US\$225.16	14 Aug 2026
Quanta ^{2,4,5,28a,6c}	2382.TW	Buy	NT\$327.50	14 Aug 2026
Renesas Electronics ^{28a}	6723.T	Buy	¥3,746	14 Aug 2026
Taiwan Semiconductor Manufacturing ^{16,28b}	2330.TW	Buy	NT\$2,395	14 Aug 2026
Vertiv Holdings Co ^{5,16,28a,6c}	VRT.N	Buy	US\$293.84	14 Aug 2026
Wistron ^{1,2,4,5,28a,6c}	3231.TW	Buy	NT\$193.50	14 Aug 2026
Wiwynn ^{5,6c}	6669.TW	Buy	NT\$6,435	14 Aug 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

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